Mehrun-Nisa Rahman

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SUMMARY

Computer Science student at BRAC University who builds AI systems and full-stack web applications end to end. Built a 3-agent vision-language pipeline that reached 97% agreement with human labels on a blind test set, and shipped payment, refund and job tracking features in a team MERN platform. Seeking internships and entry-level roles in software engineering, AI/ML or data.

EDUCATION

BRAC University, Dhaka, Bangladesh — Sep 2022 – Apr 2027 (expected)
Bachelor of Science in Computer Science

- Thesis (team of 3): Human-in-the-Loop Multi-Agent Vision-Language Framework. Supervisor: Md. Sabbir Ahmed.
- Relevant coursework: Artificial Intelligence, Computer Graphics, Database Systems, Software Engineering.

Engineering University Girls' School and College, Dhaka, Bangladesh — 2012 – 2021
Higher Secondary Certificate (HSC), Science, 2021 | GPA 5.00
Secondary School Certificate (SSC), Science, 2019 | GPA 4.44

PROJECTS

LoopLens: Human-in-the-Loop Listing-Mismatch Detection | Python, FastAPI, Gemini vision model — Oct 2026

- Built a 3-agent vision-language pipeline (Router, Photo Inspector, Judge) that checks whether an online buyer's item matches the listing, with a review app where a human confirms or overrules each verdict.
- Hand-labelled 90 real Amazon cases; reached 97% agreement with humans on a blind test set (kappa 0.93).
- Cut token use by 27% with text-first routing at no loss in accuracy; covered by 29 automated tests (pytest).

FixIt: Local Home Services Booking Platform (team project, Software Engineering) | MongoDB, Express, React, Node.js

- Integrated SSLCommerz payments (bKash, Nagad, cards) with automated refund rules and email notifications.
- Built a real-time job status tracker, a rating and review system with admin moderation, and service provider profiles.

Hotel Booking Status Classification | Python, scikit-learn, pandas

- Predicted check-out, cancellation or no-show on 119K bookings, comparing four classifiers and K-Means clustering.
- Best model (neural network) reached 83% test accuracy and 0.91 AUC on imbalanced classes.

Online Inventory and Order Management System | PHP, MySQL

- Built a marketplace with cart, checkout and payments on an 11-table relational database schema.

Traffic Rider: 3D Endless-Runner Game | Python, PyOpenGL

- Built a 3D lane-based game with obstacles, collectibles, rising difficulty and two camera views.

EXPERIENCE AND LEADERSHIP

Private Tutor (Part-time) | Dhaka, Bangladesh — 2018 – 2026 (intermittent)

- Tutored kindergarten and primary school students, planning lessons and adapting explanations to each child's level.

Executive, Human Resources | BRAC University Computer Club (BUCC) — Feb 2023 – 2025

- Supported the club's human resources team with member coordination and club events.

SKILLS AND CERTIFICATIONS

- **Programming languages:** Python, JavaScript, PHP, SQL, C
- **Machine learning and AI:** scikit-learn, pandas, NumPy, matplotlib, classification, clustering (K-Means), neural networks, model evaluation (ROC-AUC, F1, Cohen's kappa), vision-language models, multi-agent systems, prompt design
- **Web development:** FastAPI, React, Node.js, Express, MongoDB, MySQL, REST APIs, HTML, CSS, relational database design, SSLCommerz payment integration, OpenGL
- **Tools:** Git, GitHub, GitHub Actions, pytest, Jupyter, Google Colab, VS Code
- **Certifications:** AWS Student Community Day Participant, AWS Community (Credly badge), Jan 2026
- **Languages:** Bengali (native), English (fluent)